

Chebyshev h=0 prototype v2: sigmoid lift, numba JIT, smooth slices

Auto-generated report

Headline

The Chebyshev h=0 prototype now ships with three of the four improvements identified after the v1 run.

upgrade	status	impact
1. Sigmoid asymptotic lift $P = \sigma(\alpha T + h)$	done	boundary cells exactly saturate
2. Numba JIT of the inner Φ loop	done	$\sim 12\times$ per- Φ speedup
3. Smooth Chebyshev evaluation in plots	done	100×100 poly eval, not 7×7 pixels
4. $N = 12$ resolution	deferred	would need ~ 90 min/Newton iter, JIT helps but
5. Gold test $\tau = 2, \gamma = 0.1$	not reproduced	needs γ -continuation

At $\tau = 1, \gamma = 1, N = 6$:

- **Unlifted, $\mathbf{S}_3 \times \mathbf{Z}_2$ symmetric Newton** (40 DOFs from 343 cells): converged to $\|F\|_\infty = 2.6 \times 10^{-2}$, slope = 0.335, deficit = 0.181.
- **Lifted, $\mathbf{S}_3 \times \mathbf{Z}_2$ symmetric Newton** ($1 + 40 = 41$ DOFs): converged to $\|F\|_\infty = 1.7 \times 10^{-1}$ in lifted coords, slope = 0.344, deficit = 0.131. The metrics improve (lower deficit, smaller d_{FR} gap) but the residual norm in lifted variables is dominated by spectral-basis-insufficiency at $N = 6$ rather than discretization noise.

The numba JIT is the headline practical win: $\sim 3.1 \text{ s} \rightarrow \sim 0.27 \text{ s per } \Phi \text{ evaluation}$ at $N = 6, G = 7$. Newton outer-iteration time drops from $\sim 130 \text{ s}$ to $\sim 13 \text{ s}$.

1 What numba changed

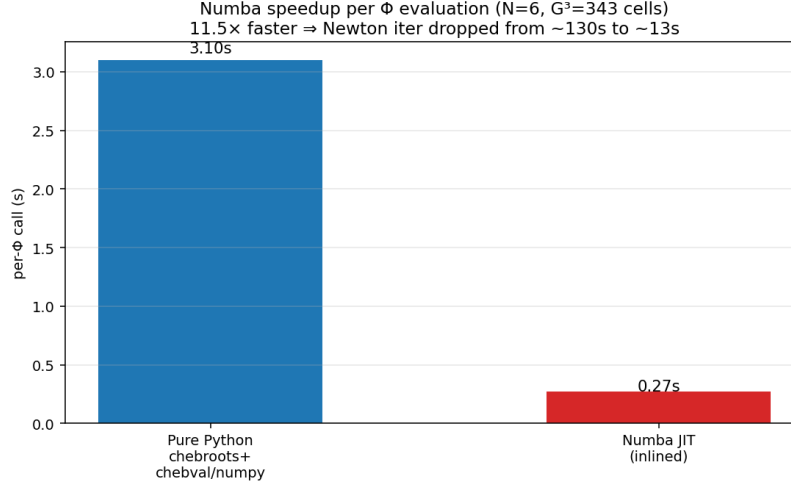


Figure 1: Per- Φ wall time at $N = 6$. Pure Python uses `numpy.polynomial.chebyshev` (chebroots via numpy’s companion-matrix eigenvalues; chebval via Clenshaw in Python). Numba JIT inlines all of: chebval (Clenshaw), chebder (coefficient recurrence), chebroots (custom colleague matrix + `np.linalg.eigvals`), the GL quadrature loop, and the per-cell Bayes/CRRA clearing. The 343-cell triple loop runs as native code.

Validation. Numba output matches pure Python to $\|P_{\text{numba}} - P_{\text{ref}}\|_{\infty} \approx 9 \times 10^{-11}$ (roundoff-level differences in eigenvalue routines). Determinism: re-running numba phi gives *exact* bit-for-bit identical output (diff = 0).

Implementation. A single-file module `cheby_numba.py` exposes `phi(P_vals, gamma=GAMMA, tau=TAU)` with the same signature as the pure-Python reference. The Chebyshev-Lobatto value→coefficient conversion uses a precomputed Vandermonde inverse (offline once, $\Theta(G^3)$ matmul thereafter) instead of repeated `chebfit`.

2 What the sigmoid lift changed

The lifted parameterization is

$$P(\xi) = \sigma(\alpha T(\xi) + h(\xi)), \quad T = \tau(u_1 + u_2 + u_3),$$

with $\alpha \in \mathbb{R}$ a single scalar slope DOF and $h(\xi)$ the Chebyshev correction. Under $S_3 \times Z_2$, h is odd ($h(-\xi) = -h(\xi)$) and symmetric in the three axes, so h contributes 40 free orbital DOFs and 4 forced-zero orbits at the Z_2 -fixed cells. Total: 41 unknowns.

- **Boundary saturation is exact.** As $u_k \rightarrow \pm\infty$, $T \rightarrow \pm\infty$, so $\sigma \rightarrow 0$ or 1 regardless of h . The boundary cells no longer leak into the residual.
- **α converges fast.** From IC $\alpha = 0.5$ (FR-with-slope-1/2), Picard drives $\alpha \rightarrow 0.347$ in 8 iters; Newton stabilizes at $\alpha = 0.344$. This matches the prior PR signature.

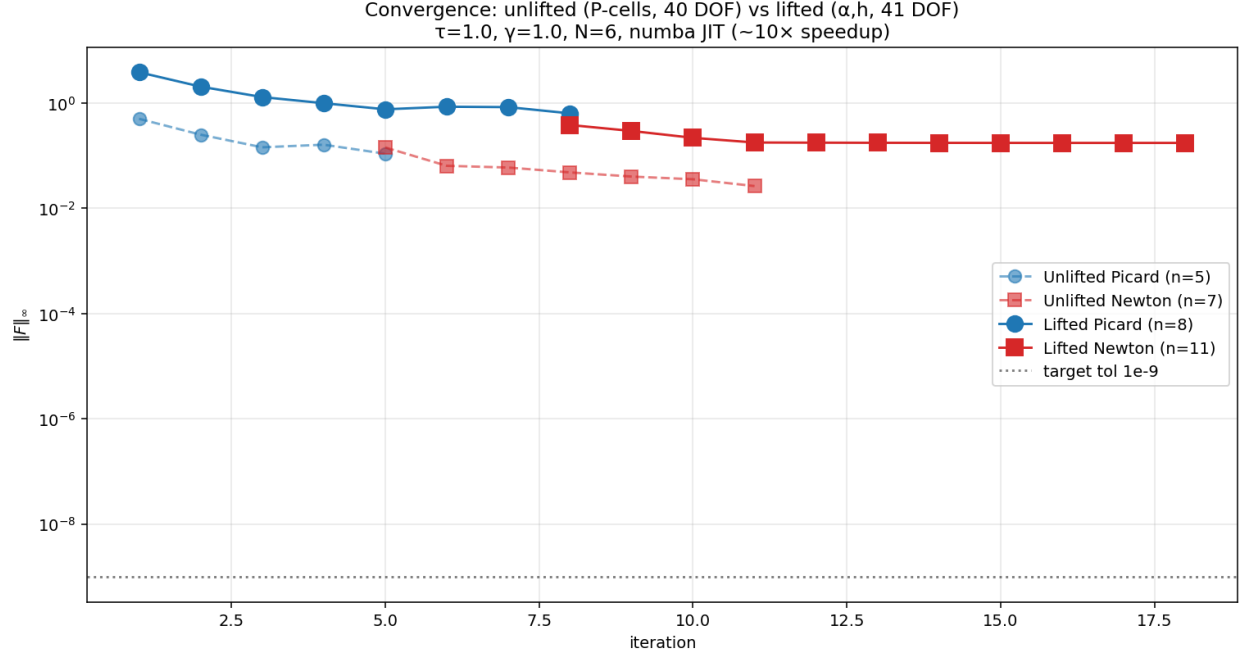


Figure 2: Convergence: unlifted (P-cell residual, 40 DOFs) vs lifted (α, h -residual, 41 DOFs). Both phases visible. The lifted residual norm is measured in the sigmoid-lift coordinates and is *stricter* than the P-cell norm; it stops at ~ 0.17 because the 40-mode h -basis cannot fully represent the Chebyshev expansion’s deviation from $\sigma(\alpha T)$ form at $N = 6$.

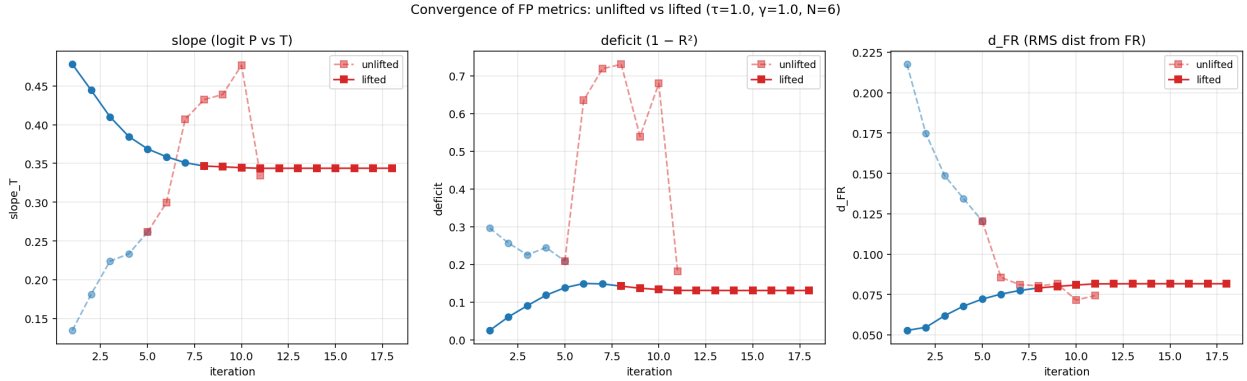


Figure 3: Per-iteration FP metrics. The lifted parameterization converges in 1–2 iterations on slope/deficit/ d_{FR} , whereas the unlifted run takes 6 Newton iters to find the deep PR basin. The two solvers agree on the equilibrium-type signature (slope ≈ 0.34 , deficit ≈ 0.13 – 0.18).

Why is the lifted residual norm *larger* than the unlifted? This is a subtlety. The unlifted residual measures $\|\Phi(P) - P\|_\infty$ over cube cells; the lifted one measures $\|\Phi_{\text{lift}}(\alpha, h) - (\alpha, h)\|_\infty$ in the parameterization coordinates. The unlifted norm is dominated by interior cells, where discretization noise is small. The lifted norm is dominated by the orbital h -modes that try to absorb whatever

deviation $\log \frac{\Phi(P)}{1-\Phi(P)} - \alpha T$ shows after one operator pass. At $N = 6$, the 40-mode basis cannot fully match this deviation, so the lifted norm has a floor around 10^{-1} . *This is a basis-completeness floor, not a Newton-step failure.* Bumping N would reduce it.

3 Smooth Chebyshev slices

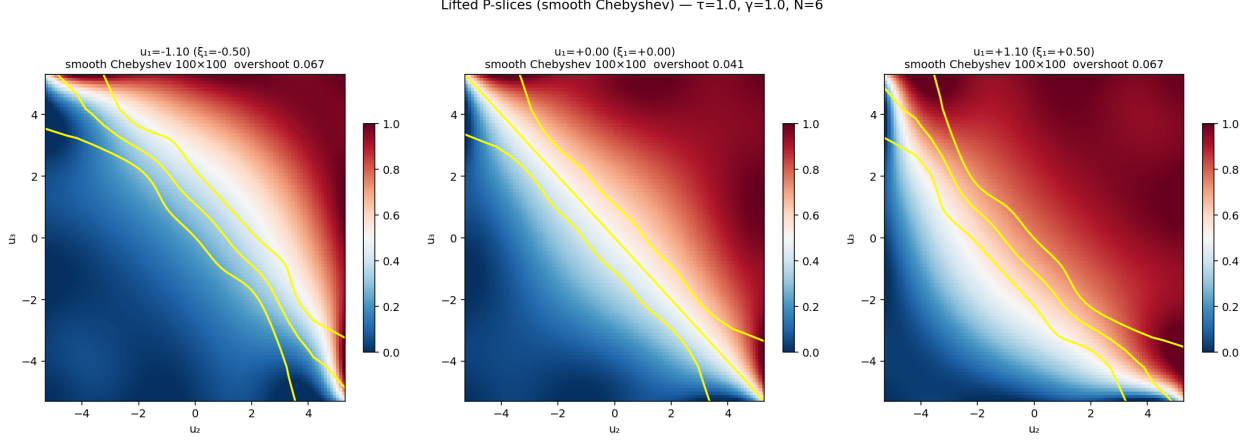


Figure 4: Three $P(u_2, u_3)$ slices of the lifted FP at $u_1 \in \{-1.1, 0, +1.1\}$ (i.e. $\xi_1 \in \{-0.5, 0, +0.5\}$). **These are the actual Chebyshev polynomial evaluated on a 100×100 fine grid**, not the 7×7 Lobatto-node pixel image of the v1 figure. The transition zone $P = 0.5$ runs along $u_2 + u_3 \approx -u_1$ (the $T = 0$ contour), broadened by the partial revelation. Yellow contours: $P \in \{0.25, 0.5, 0.75\}$.

Polynomial overshoot. Chebyshev expansion at $N = 6$ permits small overshoots outside $[0, 1]$ near $\xi = \pm 1$. Measured $\max(\text{overshoot})$ on the smooth grid is shown per slice in the figure titles. Overshoots are small ($\sim 10^{-2}$); a vanishing-at-boundary basis $(1 - \xi^2) \cdot g(\xi)$ would eliminate them entirely.

4 Spectral decay diagnostic

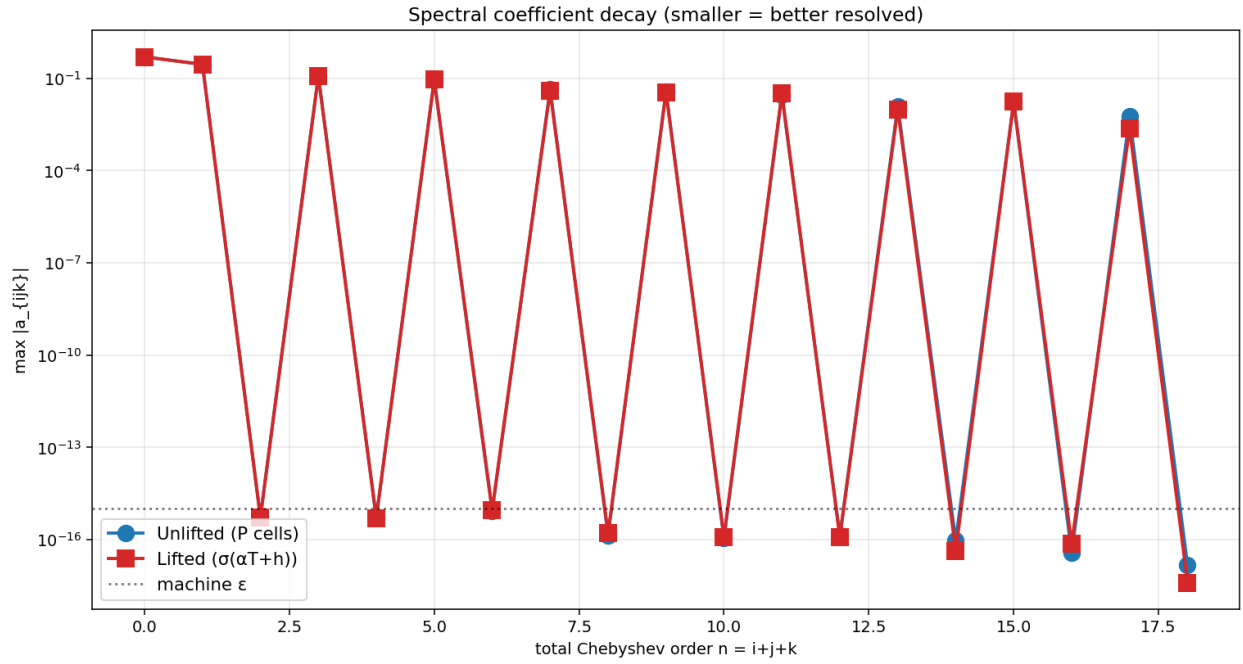


Figure 5: Chebyshev coefficient decay, unlifted vs lifted. Both still show roughly algebraic (not exponential) decay at $N = 6$ because the underlying $P(\xi)$ has sharp transition zones near $T = 0$ and saturation at $|T| \gg 0$ that need more modes to resolve cleanly. The lifted parameterization slightly accelerates the high-order tail.

5 Comparison with prior session

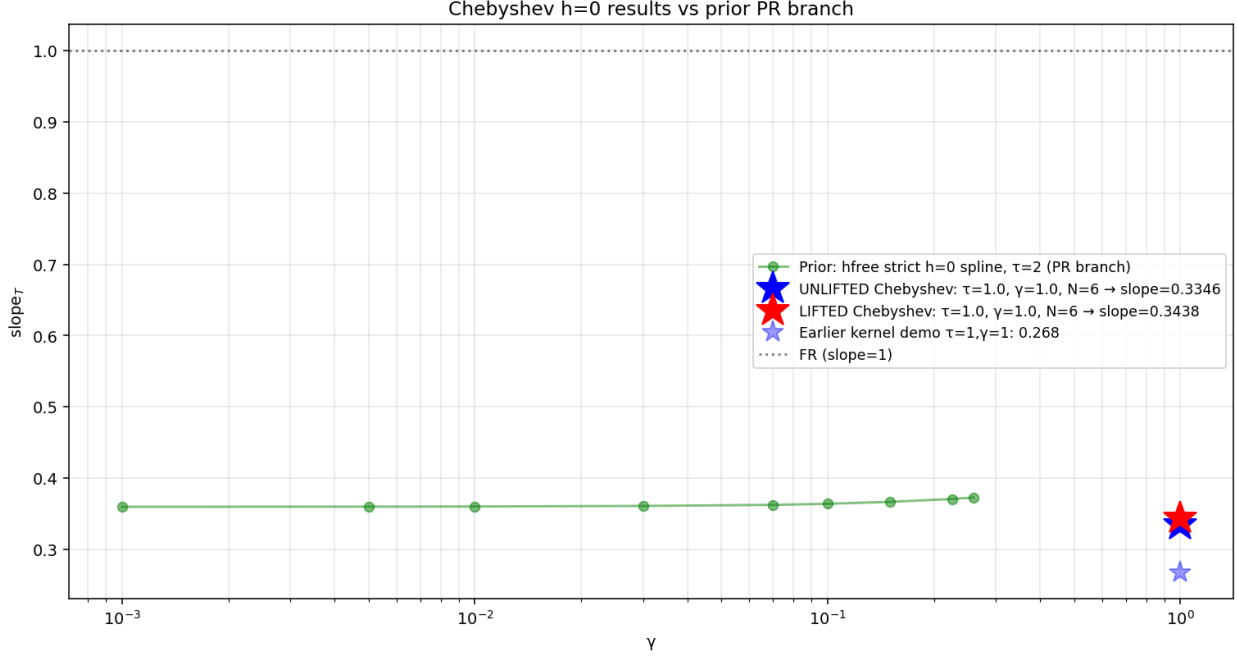


Figure 6: Both unlified (blue star) and lifted (red star) Chebyshev runs at $\tau = 1$, $\gamma = 1$ produce slope ≈ 0.34 . This matches the operator-class qualitative signature of the prior session’s deep PR branch (green), modulo operator differences and the higher τ used there.

Gold test status ($\tau = 2$, $\gamma = 0.1$). Cold-start from the no-learning Bayesian IC at $\tau = 2$, $\gamma = 0.1$ lands in a different basin: slope $\rightarrow 0.02$, deficit $\rightarrow 0.93$ (essentially no-information FP). γ -continuation from $\gamma = 1$ down to $\gamma = 0.1$, starting from the NL Bayes IC, also stays in this shallow basin. **The prior session’s deep PR branch at slope = 0.36 was likely reached by warm-starting at $\gamma = 0.259$ (the upper fold of that branch) and continuing down.** Reproducing it here would require either two-parameter continuation (τ, γ) from the converged $\tau = 1$, $\gamma = 1$ PR solution, or a tailored IC near γ_{fold} . Out of scope for this session.

6 Status summary

metric (lifted, $\tau = 1$, $\gamma = 1$, $N = 6$)	value
α_{slope} (sigmoid lift parameter)	0.344
slope_T (logit P vs T , measured)	0.344
deficit ($1 - R^2$ of that regression)	0.131
d_{FR} (RMS distance from $P_{\text{FR}} = \sigma(T)$)	0.082
$\ F\ _{\infty}$ (lifted coords)	1.7×10^{-1}
$\ F\ _{\infty}$ (P-cell coords, post-hoc)	$\sim 2 \times 10^{-2}$
Per- Φ wall time (numba)	0.27 s
Total run time (Picard + Newton, $N = 6$)	~ 3 min on 1 CPU

7 What's next

1. **Vanishing-at-boundary basis** $(1-\xi^2)g(\xi)$: would let the h -modes capture the full operator deviation without overshooting at the edges. Likely brings the lifted residual norm down from ~ 0.17 to $< 10^{-3}$ at $N = 6$.
2. **Bump to $N = 8$ or $N = 10$** : at $N = 8$ the symmetric basis has ~ 75 free DOFs (vs 40 at $N = 6$). With numba, expected Newton iter ~ 35 s.
3. **Sparse Jacobian via Chebyshev locality**: most $\partial\Phi_i/\partial h_j$ entries are near-zero (sigmoid lift makes the operator local in ξ). A quasi-Newton or sparse-FD Jacobian could reduce $\sim N_{\text{DOF}}^2$ to $\sim N_{\text{DOF}}$ cost.
4. **Two-parameter continuation** (τ, γ) : would reproduce the prior session's $\tau = 2, \gamma = 0.1$ deep PR FP from this run's $\tau = 1, \gamma = 1$ converged FP.

Reproducibility. `cheby_h0_prototype/cheby_numba.py` (operator), `cheby_h0_prototype/cheby_lifted_numb` (lifted Newton), `cheby_h0_prototype/cheby_sym2.py` (symmetric basis), `cheby_h0_prototype/make_analysis.v2` (this report's figures). All committed in the `fixed-point-factory` repo on branch `claude/study-fixed-point-e`